

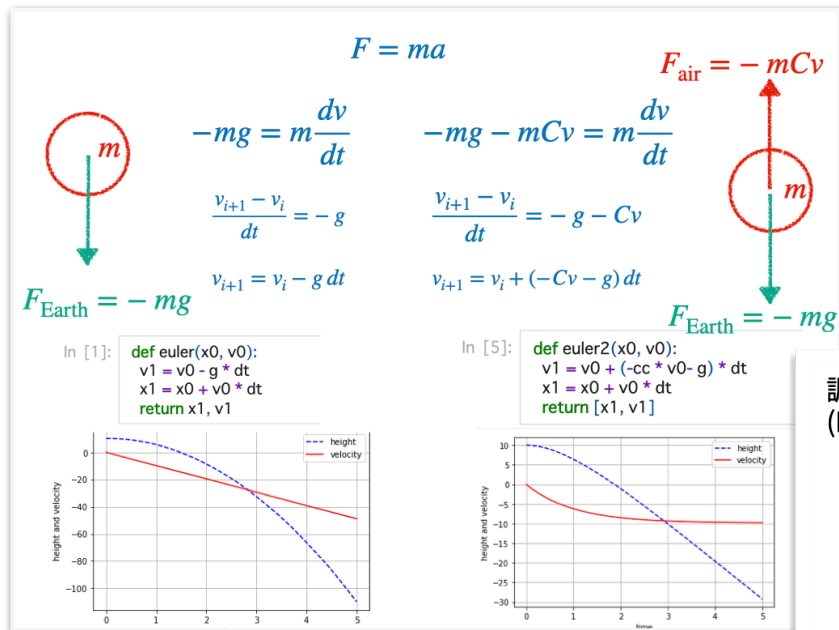
重力のイメージ

keyword: $F=ma$, free fall, air resistance,

Kepler, Harmonic law

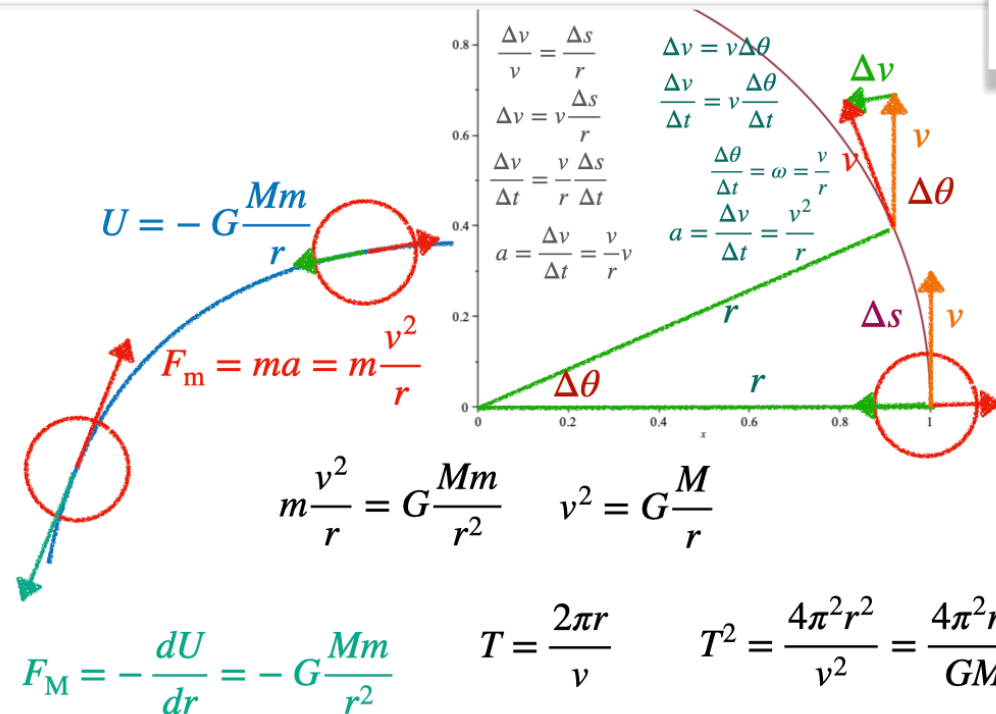
source: kepler_mahmonic, SolarSystem.mw, d1_python_ode.ipynb

daddygongon 26/5/31



調和の法則 Law of Harmonies (Kepler's third law of planetary motion)

惑星	公転周期(T)	軌道長半径(r)	T^2/r^3 by Copilot	T^2/r^3 by you
水星	0.241	0.387	0.037	
金星	0.615	0.723	0.078	
地球	1.000	1.000	1.000	1.000
火星	1.881	1.524	0.296	
木星	11.86	5.203	1.000	
土星	29.46	9.537	0.930	
天王星	84.01	19.191	1.000	
海王星	164.8	30.07	1.000	



$$\vec{r} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos(\omega t) \\ r \sin(\omega t) \end{pmatrix}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} -r\omega \sin(\omega t) \\ r\omega \cos(\omega t) \end{pmatrix}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \begin{pmatrix} -r\omega^2 \cos(\omega t) \\ -r\omega^2 \sin(\omega t) \end{pmatrix}$$

$$\vec{a} = -\omega^2 \begin{pmatrix} r \cos(\omega t) \\ r \sin(\omega t) \end{pmatrix} = -\omega^2 \vec{r}$$

$$v = r\omega$$

$$a = r \left(\frac{v}{r} \right)^2 = \frac{v^2}{r}$$

